

GameDVR Fullscreen-Optimization Registry Keys

Windows maintains two registry keys under `HKCU\System\GameConfigStore` – **GameDVR_FSEBehaviorMode** and **GameDVR_FSEBehavior** – that control the “Fullscreen Optimizations” feature (sometimes called the FSE (fullscreen-exclusive) mode). In general, these are global settings affecting how full-screen games/applications are handled (the Windows “game bar”/DVD recording subsystem). In particular, setting them can force all games to use (or not use) fullscreen optimizations. User reports and Microsoft community answers explain their meaning and valid values as follows:

- **GameDVR_FSEBehaviorMode** – A global toggle for fullscreen optimizations. By default its value is `0` (Fullscreen Optimizations **enabled**). Changing it to `2` disables FSO globally. In other words, `0` = fullscreen optimizations ON (default), `2` = OFF ¹. (Some sources note only 0 and 2 are used; there is no commonly documented meaning for 1.) This DWORD lives in HKCU and thus applies to the current user account system-wide – it is *not* per-game. For example, a Microsoft Q&A reply confirms that a `0` value means FSO is enabled and changing it to `2` turns it off for all games ¹. On Windows 10 (through 1809/1903) this toggle historically worked to disable FSO globally; in Windows 11 the key still exists but Microsoft has effectively removed any official global-disable support, so its effect may be limited.
- **GameDVR_FSEBehavior** – An earlier/alternate global flag for fullscreen optimizations. This key also resides in `HKCU\System\GameConfigStore`. Community experts report the same 0/2 scheme: setting `0` = FSO enabled, `2` = FSO disabled ². (One advanced tweak guide shows the original/default value as 1 and sets it to 2 to disable ³, implying `1` may be a legacy or default state.) Like FSEBehaviorMode, FSEBehavior is a global toggle for FSO – it is *not* inherently per-app. For example, a WindowsReport article on the May 2019 update advises setting `GameDVR_FSEBehavior=2` along with the other keys to restore classic fullscreen mode ⁴. In practice, many tweak guides recommended setting **both** `FSEBehaviorMode=2` and `FSEBehavior=2` (and related keys) to fully disable enhanced fullscreen mode ⁴ ².

Key Values & Meaning

The clear consensus from forum and Q&A sources is:

- **0 (enabled):** Fullscreen optimizations are enabled (the new “enhanced fullscreen” mode is active).
- **2 (disabled):** Fullscreen optimizations are turned off (Windows will use legacy exclusive fullscreen for games).

Both keys appear to use this same interpretation. For example, a Superuser answer explicitly states that the default `0` means FSO is on, and changing it to `2` turns it off system-wide ¹. Likewise, a Guru3D forum expert confirms that `GameDVR_FSEBehavior 0 = Enable`, `2 = Disable` ². (The value `1` is seldom mentioned in practice – one walkthrough shows FSEBehavior’s default as 1 – but only 0 vs 2 are documented.)

Global vs. Per-App Scope

Both `GameDVR_FSEBehaviorMode` and `GameDVR_FSEBehavior` are in `HKEY_CURRENT_USER\System\GameConfigStore`. That makes them **global (user-wide) settings**, not per-game. In other words, they set the default fullscreen-optimization policy for all games in that user account. In practice, however, *per-game overrides* exist: Windows also stores game-specific FSO settings under subkeys of `HKCU\System\GameConfigStore\Children`. For example, when you tick “Disable Fullscreen Optimizations” in a game’s Compatibility tab, Windows writes flags under that game’s entry in `GameConfigStore\Children`. The keys `HonorUserFSEBehaviorMode` and `DXGIHonorFSEWindowsCompatible` control how those per-app flags are applied. Broadly, setting **GameDVR_HonorUserFSEBehaviorMode = 1** tells Windows to honor the per-game compatibility setting (i.e. if the user disabled FSO for an exe, that choice takes effect). If `HonorUserFSEBehaviorMode = 0` (the default), then the global key overrides the per-app flag. (For example, the May 2019 update workaround suggests setting `HonorUserFSEBehaviorMode=1` along with disabling FSEBehavior/FSEBehaviorMode ⁴.) In short, FSEBehaviorMode/Behavior set the *default* FSO policy; individual games’ toggles (in the Children subkeys) can override it if HonorUser... is enabled.

Windows 10 vs. Windows 11

These registry keys first appeared in Windows 10 (around the Creators Update era) to give a global control over the new Fullscreen Optimizations feature. In Windows 10 versions up through 1803, setting these keys could force all games in or out of enhanced fullscreen. **However, starting with Windows 10 1809/1903, Microsoft removed the ability to globally disable FSO.** Community reports note that after the October 2018 update, the system “defaults to on in all games, regardless of the `GameDVR_FSEBehavior` value” ⁵. In other words, Windows began ignoring the global registry hacks. Windows 11 continues this policy: fullscreen optimizations still exist (and by default apply to Win11 games), but global-disable keys may be ignored or reset by the OS. In practice, the Microsoft-recommended approach on Win10/11 is to disable FSO per-game via the Compatibility tab or Xbox Game Bar settings, rather than rely on `GameDVR_FSEBehaviorMode`.

Caveats & Related Keys

- **HonorUserFSEBehaviorMode:** As noted, this companion key (0 or 1) determines whether the user’s per-game FSO toggle is respected. Setting `HonorUserFSEBehaviorMode = 1` allows the Compatibility tab setting for FSO to take precedence, while `0` enforces the global mode ⁴. Some tweak guides set this to 1 when disabling FSO globally, to ensure consistency. However, users report that Windows often resets `HonorUserFSEBehaviorMode` (and the related `DXGIHonorFSEWindowsCompatible` flag) back to 0 on reboot or update ⁶, so these are not reliable persistent controls.
- **DXGIHonorFSEWindowsCompatible:** This key (0 or 1) appears in many registry fixes. Setting it to `1` is said to disable another compatibility mode that can force “enhanced fullscreen” in certain games. (It governs how DirectX’s DXGI handles fullscreen compatibility.) It is not part of the official documentation, but is routinely paired with the above keys in community fixes ⁴.

- **GameDVR_Enabled / AllowGameDVR:** These are separate keys that turn the Game Bar/DVR features on or off entirely (global GameDVR capture). They do not directly control FSO, but users disabling fullscreen optimizations often also disable GameDVR. For completeness: setting `HKCU\System\GameConfigStore\GameDVR_Enabled=0` and the policy `HKLM\...\GameDVR\AllowGameDVR=0` will turn off GameDVR ³.
- **Global vs. Local Effects:** Because Windows 10/11 natively handle FSO per-application, these global registry toggles are best thought of as default/baseline settings. Even with `FSEBehaviorMode=2`, an app might still use FSO if its compatibility setting re-enables it. In newer Windows, group policy or system updates may override these keys entirely.

Sources: Microsoft community and tech-forum discussions give the clearest insight. A Superuser answer confirms the meaning of `FSEBehaviorMode` (0 = on, 2 = off) and notes it applies system-wide ¹. A Windows gaming forum post similarly confirms `FSEBehavior` (0=Enable, 2=Disable) ². A WindowsReport article (May 2019) provides the recommended registry values for disabling enhanced fullscreen (setting both keys to 2, plus `HonorUserFSEBehaviorMode=1`) ⁴. Note that all these are community-documented tweaks; official Microsoft docs do not publish these keys, so one should test carefully before relying on them.

¹ How to stop taskbar from popping up on FullScreen games in Windows 10? - Super User

<https://superuser.com/questions/1371956/how-to-stop-taskbar-from-popping-up-on-fullscreen-games-in-windows-10>

² Fix game stutter on Win 10 1703-1809 | Page 6 | guru3D Forums

<https://forums.guru3d.com/threads/fix-game-stutter-on-win-10-1703-1809.420251/page-6>

³ Disable GameDVR - WinUtil Documentation

<https://christitustech.github.io/winutil/dev/tweaks/Essential-Tweaks/DVR/>

⁴ Windows 10 v1903 update prevents games from running in classic fullscreen

<https://windowsreport.com/windows-10-version-1903-gaming/>

⁵ Can't disable 'Fullscreen optimizations' globally after October 1809 update : r/Windows10

https://www.reddit.com/r/Windows10/comments/9ocux4/cant_disable_fullscreen_optimizations_globally/

⁶ Question - How to make sure Fullscreen optimization is disabled? | Tom's Hardware Forum

<https://forums.tomshardware.com/threads/how-to-make-sure-fullscreen-optimization-is-disabled.3658855/>